

Hotel Booking Cancellation Prediction Using Machine Learning

Machine Learning Assessment Report — Hotel Booking Demand Dataset

1. Abstract

This project develops machine learning models to predict whether a hotel booking will be cancelled. The Hotel Booking Demand dataset was cleaned, explored, transformed through feature engineering, and used to train four baseline classification models: Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM). A GridSearchCV-based tuning process was then applied to the Random Forest model. Model performance was evaluated using accuracy, precision, recall, and F1 score. On the reported test set, the tuned Random Forest achieved 0.8401 accuracy and a 0.6722 F1 score.

2. Problem Statement

Hotel cancellations can affect room availability, revenue planning, and operational decisions. The objective is to build a binary classification system for **is_cancelled**, where 0 represents a booking that was not cancelled and 1 represents a cancelled booking.

3. Dataset and Preprocessing

The project uses the Hotel Booking Demand dataset containing reservation records from city and resort hotels. The workflow includes missing-value and duplicate inspection, data cleaning, exploratory data analysis, feature engineering, and preprocessing of numerical and categorical variables.

Three derived features were created: **total_guests**, **total_nights**, and **total_stay_cost**. To reduce target leakage and remove fields treated as unnecessary identifiers, **reservation_status**, **reservation_status_date**, **agent**, and **company** were removed before model training. An 80/20 stratified train-test split was used. Numerical variables were median-imputed and standardized, while categorical variables were most-frequent imputed and one-hot encoded.

4. Exploratory Data Analysis

The correlation matrix below shows the relationships among the numerical variables used during exploration. The target variable is included to inspect its linear association with numerical features.

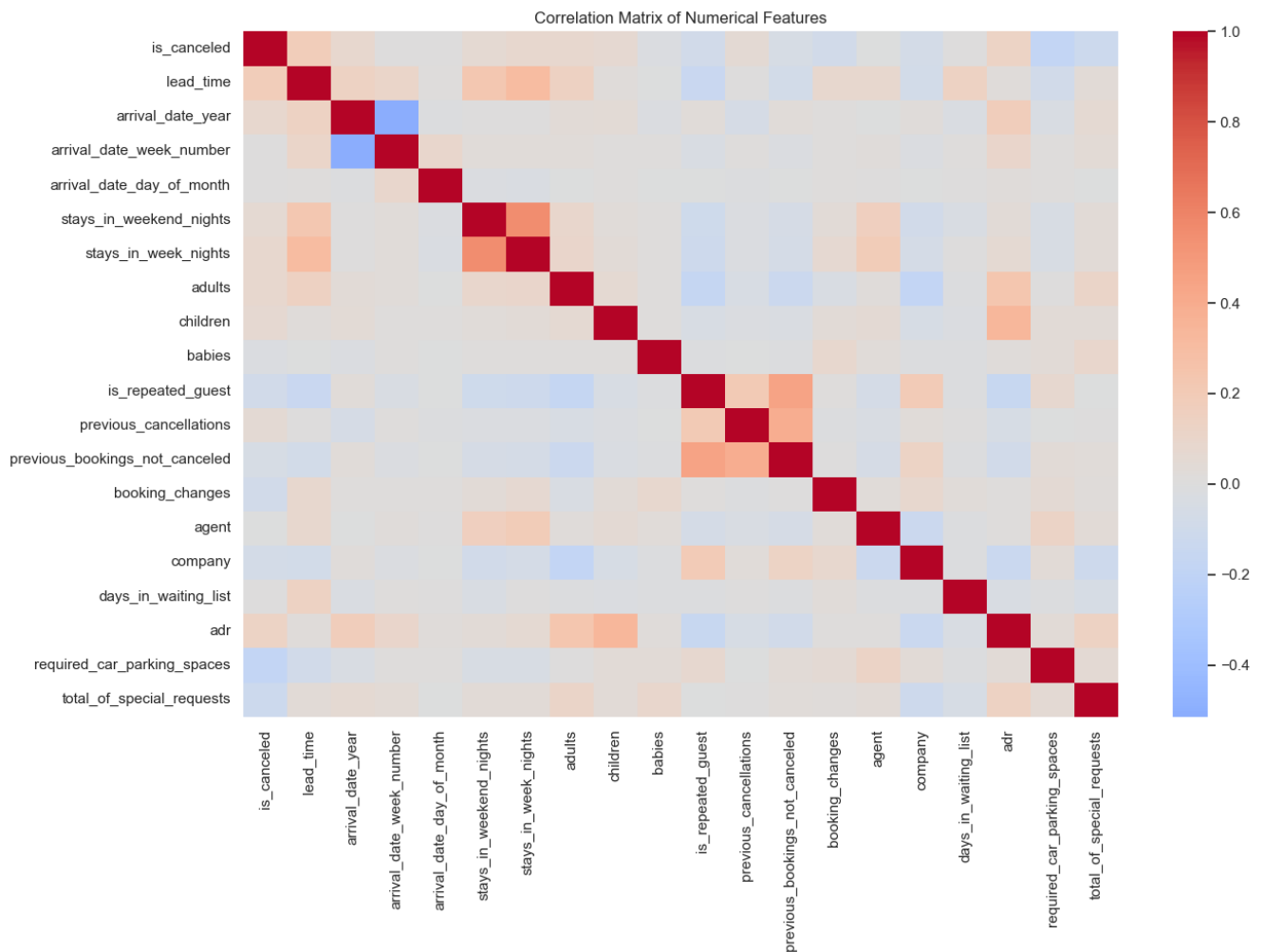


Figure 1. Correlation matrix of numerical features.

5. Models Evaluated

Model	Purpose
Logistic Regression	Linear baseline classification model
Decision Tree	Non-linear tree-based baseline
Random Forest	Ensemble tree-based classifier
SVM	Support Vector Machine baseline
Tuned Random Forest	Random Forest optimized using GridSearchCV

6. Confusion Matrix Analysis

The confusion matrices show the counts of correct and incorrect predictions for the four baseline models. For example, the Random Forest correctly classified 11,788 non-cancelled bookings and 2,819 cancelled bookings, while 829 non-cancelled bookings and 1,967 cancelled bookings were misclassified.

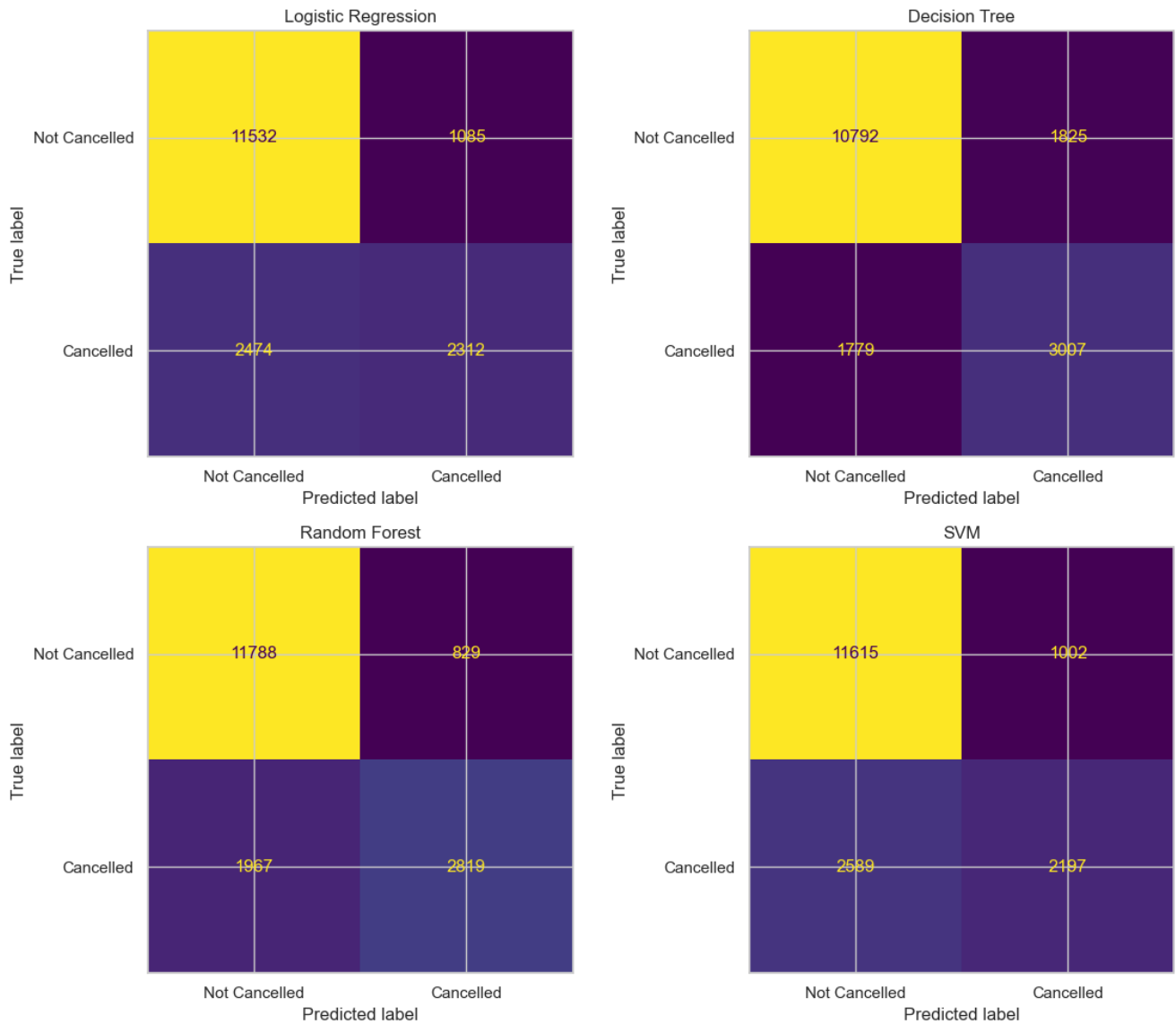


Figure 2. Confusion matrices for the four baseline models.

7. Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.7955	0.6806	0.4831	0.5651
Decision Tree	0.7929	0.6223	0.6283	0.6253
Random Forest	0.8393	0.7728	0.5890	0.6685
SVM	0.7937	0.6868	0.4590	0.5503
Tuned Random Forest	0.8401	0.7708	0.5959	0.6722

The tuned Random Forest increased test accuracy from 0.8393 to 0.8401 and F1 score from 0.6685 to 0.6722 relative to the original Random Forest. Its precision was 0.7708 and recall was 0.5959.

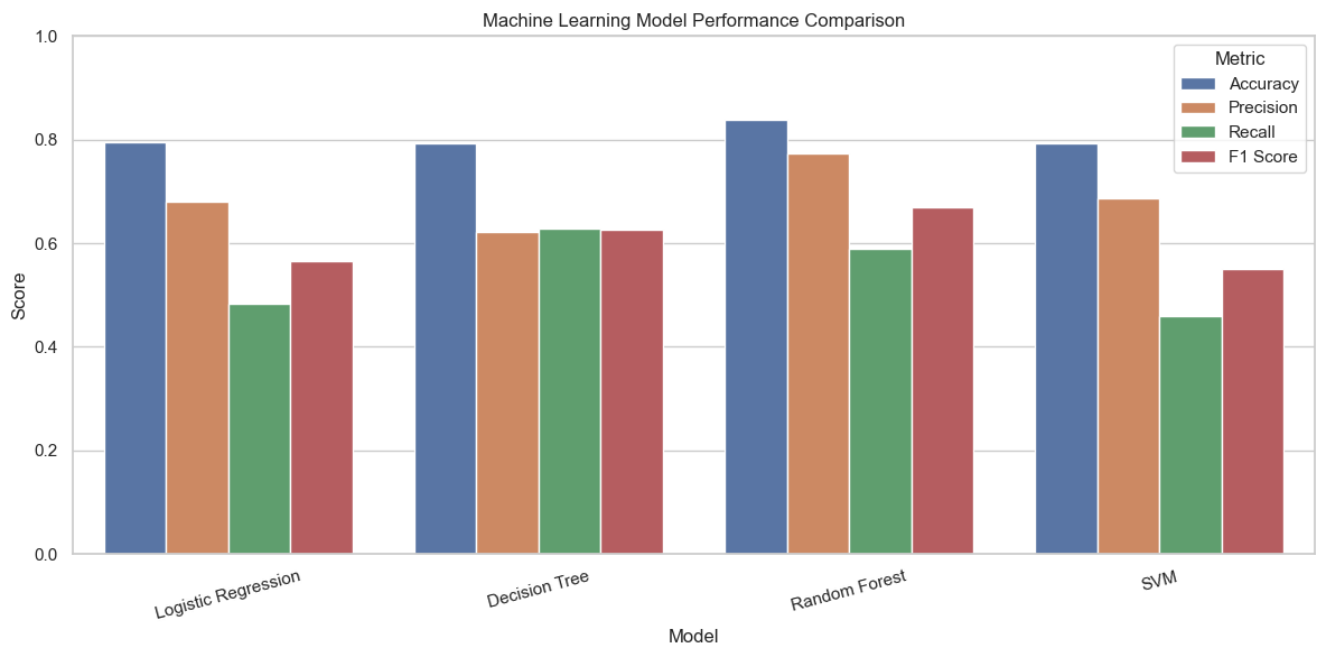


Figure 3. Baseline model performance comparison.

8. Classification Report — Tuned Random Forest

For the tuned Random Forest, the supplied classification report gives class-0 precision of 0.86, recall of 0.93, and F1 score of 0.89. For the cancellation class (class 1), precision is 0.77, recall is 0.60, and F1 score is 0.67. The overall test accuracy is 0.84.

9. Feature Importance

Feature	Importance
num__lead_time	0.110873
num__total_stay_cost	0.072963
num__adr	0.072675
num__arrival_date_day_of_month	0.058408
num__total_of_special_requests	0.054437
num__arrival_date_week_number	0.052803
cat__country_PRT	0.040808
num__total_nights	0.033432
num__stays_in_week_nights	0.031792
cat__market_segment_Online TA	0.025404
num__required_car_parking_spaces	0.025180
num__arrival_date_year	0.024545
num__stays_in_weekend_nights	0.022545

Lead time has the largest reported feature importance (0.110873), followed by total_stay_cost (0.072963) and adr (0.072675). These are model-derived importance values and should not be interpreted as evidence of causation.

10. Model Tuning

Random Forest hyperparameters were searched using GridSearchCV with 5-fold cross-validation and F1 scoring. The search considered n_estimators values of 100 and 200, max_depth values of None, 10, and 20, min_samples_split values of 2 and 5, and min_samples_leaf values of 1 and 2. The supplied output contains the tuned model's test performance, but not the exact best-parameter combination or best cross-validation F1 value, so those values are not stated here.

11. Conclusion

The experiment evaluated four baseline classifiers and a tuned Random Forest for hotel booking cancellation prediction. On the reported test set, the tuned Random Forest achieved 84.01% accuracy and a 0.6722 F1 score. The original Random Forest achieved 83.93% accuracy and a 0.6685 F1 score. Logistic Regression, Decision Tree, and SVM produced F1 scores of 0.5651, 0.6253, and 0.5503 respectively. For the cancellation class, the tuned Random Forest achieved 0.60 recall, indicating that some cancelled bookings remained difficult to identify. The results provide a quantitative comparison of the selected approaches and show the performance obtained from the tuned ensemble model on this dataset.

12. Limitations and Future Work

Future work could investigate decision-threshold tuning, class-imbalance strategies, probability calibration, additional feature engineering, and broader hyperparameter optimization. Evaluation on an independent holdout dataset or later-period bookings could also provide additional evidence about generalization.

13. Final Result Summary

Metric	Tuned Random Forest
Accuracy	0.840143
Precision	0.770811
Recall	0.595905

F1 Score	0.672166
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